

Examples:

```
In[*]:= (*We establish the multiplication for SW. We
        set the variable l to be sw. The bar denotes inverse. *)
vars = {s, w, l};
mi,j→k[Z_] := Expand[Z] /. {si sj → 1, wi wj → 0, si wj → lk, wi sj → -lk, li sj → -wk, si lj → wk,
        li wj → 0, wi lj → 0, li lj → 0, x-i /; MemberQ[vars, x] ⇒ xk, x-j /; MemberQ[vars, x] ⇒ xk}}
```

```
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        li wj → 0, wi lj → 0, li lj → 0, x-i /; MemberQ[vars, x] ⇒ xk, x-j /; MemberQ[vars, x] ⇒ xk}}
```

```
In[*]:= (* We start by declaring the elements R and κ: *)
κi := λ1 + λ2 si + λ3 wi + λ4 li;
κ̄i := λ̄1 + λ̄2 si + λ̄3 wi + λ̄4 li;
Ri,j := μ1,1 + μ1,2 si + μ1,3 wi + μ1,4 li + μ2,1 sj + μ2,2 si sj + μ2,3 wi sj + μ2,4 li sj +
        μ3,1 wj + μ3,2 si wj + μ3,3 wi wj + μ3,4 li wj + μ4,1 lj + μ4,2 si lj + μ4,3 wi lj + μ4,4 li lj
R̄i,j := μ̄1,1 + μ̄1,2 si + μ̄1,3 wi + μ̄1,4 li + μ̄2,1 sj + μ̄2,2 si sj + μ̄2,3 wi sj + μ̄2,4 li sj +
        μ̄3,1 wj + μ̄3,2 si wj + μ̄3,3 wi wj + μ̄3,4 li wj + μ̄4,1 lj + μ̄4,2 si lj + μ̄4,3 wi lj + μ̄4,4 li lj
(*We also implement the element v :*)
vi := R1,3 κ̄2 // m1,2→1 // m1,3→i
v̄i := R̄3,1 κ̄2 // m1,2→1 // m1,3→i
```

(*To change from one test to another, please exit the kernel.*)

TEST 1:

```
In[*]:= μ1,3 = 0; μ2,4 = 1; μ3,1 = 1; λ1 = 0; μ4,4 = 1; μ4,1 = 1; λ2 = -1; λ3 = -1; λ4 = -1; μ1,2 = 0;
        μ1,4 = 0; μ2,1 = 1; μ2,2 = 1; μ2,3 = 1; μ3,2 = 1; μ3,3 = 1; μ3,4 = 1; μ4,2 = 1; μ4,3 = 1;
```

TEST 2:

```
In[*]:= λ1 = 1; λ2 = 0; λ3 =  $\frac{2 \mu_{2,3}}{\mu_{1,1}}$ ; λ4 =  $-\frac{2 \mu_{2,3}}{\mu_{1,1}}$ ; μ1,2 = 0; μ2,1 = 0; μ2,2 = 0;
        μ1,3 = -μ1,4; μ2,4 = -μ2,3; μ3,1 = μ1,4; μ3,2 = -μ2,3; μ4,1 = -μ1,4;
        μ4,2 = μ2,3; μ4,3 =  $\frac{2 \mu_{1,4}^2 - 2 \mu_{2,3}^2 - \mu_{1,1} \mu_{3,4}}{\mu_{1,1}}$ ; μ4,4 =  $\frac{-2 \mu_{1,4}^2 + 2 \mu_{2,3}^2 - \mu_{1,1} \mu_{3,3}}{\mu_{1,1}}$ ;
```

TEST 3:

```
In[*]:= λ1 = 1; λ2 = 0; μ1,2 = 0; μ2,1 = 0; μ2,2 = 0; λ3 =  $\frac{2 \mu_{2,3}}{\mu_{1,1}}$ ; λ4 =  $-\frac{2 \mu_{2,3}}{\mu_{1,1}}$ ;
        μ1,3 = -μ1,4; μ2,4 = -μ2,3; μ3,1 = μ1,4; μ3,2 = -μ2,3; μ4,1 = -μ1,4;
        μ4,2 = μ2,3; μ4,3 =  $\frac{2 \mu_{1,4}^2 - 2 \mu_{2,3}^2 - \mu_{1,1} \mu_{3,4}}{\mu_{1,1}}$ ; μ4,4 =  $\frac{-2 \mu_{1,4}^2 + 2 \mu_{2,3}^2 - \mu_{1,1} \mu_{3,3}}{\mu_{1,1}}$ ;
```

Checking that each of these examples indeed defines an XC-structure on SW:

```

In[ ]:= 
$$\left( \bar{R}_{1,2} R_{3,4} // m_{1,3 \rightarrow 1} // m_{2,4 \rightarrow 2} \right) - 1 // \text{Simplify}$$


$$\left( \left( R_{1,2} R_{4,3} R_{5,6} // m_{1,4 \rightarrow 1} // m_{2,5 \rightarrow 2} // m_{3,6 \rightarrow 3} \right) - \left( R_{1,6} R_{2,3} R_{4,5} // m_{1,4 \rightarrow 1} // m_{2,5 \rightarrow 2} // m_{3,6 \rightarrow 3} \right) \right)$$


$$\left( \left( \kappa_1 \kappa_2 R_{3,4} \bar{\kappa}_5 \bar{\kappa}_6 // m_{1,3 \rightarrow 1} // m_{1,5 \rightarrow 1} // m_{2,4 \rightarrow 2} // m_{2,6 \rightarrow 2} \right) - R_{1,2} \right) // \text{Simplify}$$


$$\left( \left( \kappa_1 \kappa_2 \bar{R}_{3,4} \bar{\kappa}_5 \bar{\kappa}_6 // m_{1,3 \rightarrow 1} // m_{1,5 \rightarrow 1} // m_{2,4 \rightarrow 2} // m_{2,6 \rightarrow 2} \right) - \bar{R}_{1,2} \right) // \text{Simplify}$$


$$\left( \left( \bar{R}_{1,4} R_{3,2} \kappa_6 // m_{1,3 \rightarrow 1} // m_{2,6 \rightarrow 2} // m_{2,4 \rightarrow 2} \right) - \kappa_2 \right) // \text{Expand} // \text{Simplify}$$


$$\left( \left( \bar{R}_{5,1} R_{2,3} \bar{\kappa}_4 // m_{1,3 \rightarrow 1} // m_{2,4 \rightarrow 2} // m_{2,5 \rightarrow 2} \right) - \bar{\kappa}_2 \right) // \text{Expand} // \text{Simplify}$$


$$\left( \kappa_1 \bar{\kappa}_2 // m_{1,2 \rightarrow 1} \right) - 1 // \text{Simplify} // \text{Expand}$$


$$\left( \left( R_{1,3} \bar{\kappa}_2 // m_{1,2 \rightarrow 1} // m_{1,3 \rightarrow 1} \right) - \left( R_{3,1} \kappa_2 // m_{1,2 \rightarrow 1} // m_{1,3 \rightarrow 1} \right) \right) // \text{Expand}$$


```

Out[]:=

0

Out[]:=

0

Out[]:=

0

Out[]:=

0

Out[]:=

0

Out[]:=

0

Out[]:=

0

Out[]:=

0

```

In[ ]:= Once[AppendTo[$Path, FileNameJoin[{"Users/jorge/MEGA/PhD/MathematicaNotebooks/KnotTheory"}]]];
<< KnotTheory` ;
];

```

Get: Cannot open KnotTheory`.

```

In[ ]:= RVK::usage =
"RVK[xs, rots] represents a Rotational Virtual Knot with a list of n Xp/Xm crossings
xs and a length 2n list of rotation numbers rots. Crossing
sites are indexed 1 through 2n, and rots[[k]] is the rotation
between site k-1 and site k. RVK is also a casting operator
converting to the RVK presentation from other knot presentations.";

```

```

In[*]:= RVK[pd_PD] := Module[{n, xs, x, rots, front = {0}, k},
  n = Length@pd; rots = Table[0, {2 n}];
  xs = Cases[pd, x_X :-> { Xp[x[[4]], x[[1]] PositiveQ@x,
                           Xm[x[[2]], x[[1]]      True      }];
  For[k = 0, k < 2 n, ++k, If[k == 0 ∨ FreeQ[front, -k],
    front = Flatten@Replace[front, k -> (xs /. {
      Xp[k + 1, l_] | Xm[l_, k + 1] :-> {l, k + 1, 1 - l},
      Xp[l_, k + 1] | Xm[k + 1, l_] :-> (++rots[[l]]; {1 - l, k + 1, l}),
      _Xp | _Xm :-> {}
    }), {1}],
    Cases[front, k | -k] /. {k, -k} :-> --rots[[k + 1]]];
  RVK[xs, rots];
RVK[K_] := RVK[PD[K]];

```

```

In[*]:= rot[i_, 0] := 1;
rot[i_, n_] := rot[i, n];
rot[i_, n_] := Module[{j},
  rot[i, n] = If[n > 0, rot[i, n - 1] κj, rot[i, n + 1] κ̄j] // mi,j→i;

```

```

In[*]:= Width[pd_PD] := Max[Length/@FoldList[Complement[#1 ∪ #2, #1 ∩ #2] &, {}, List@@List@@@pd]]

```

```

In[*]:= ThinPosition[K_] := Module[{todo, done, pd, cc},
  todo = List@@PD@K; done = {}; pd = PD[];
  While[todo != {},
    AppendTo[pd, cc = RandomChoice@MaximalBy[todo, Length[done ∩ List@@#] &]];
    todo = DeleteCases[todo, cc];
    done = done ∪ List@@cc];
  pd];
ThinPosition[K_, n_] := First@MinimalBy[Table[ThinPosition[K], n], Width];

```

```

In[ ]:= Z[K_] := Z[RVK@ThinPosition[K, 100]];
Z[rvk_RVK] := Monitor[Module[{ζ, done, st, cc, χ, i, j, k},
  ζ = 1; done = {}; st = Range[2 Length[rvk[[1]]]; $M = {};
  Do[AppendTo[$M, cc];
    {i, j} = List @@ cc;
    χ = (cc /. {_Xp :-> Ri,j v0, _Xm :-> Ri,j v0}) // mj,0→j;
    Do[χ = (rot[0, rvk[[2, k]]] χ) // m0,k→k, {k, {i, j}}];
    ζ *= χ;
    Do[
      If[MemberQ[done, k + 1], ζ = ζ // mk,k+1→k; st = st /. k + 1 -> k];
      If[MemberQ[done, k - 1], ζ = ζ // mst[[k-1],k→st[[k-1]]; st = st /. k -> st[[k-1]],
        {k, {i, j}}];
      done = done ∪ {i, j},
      {cc, rvk[[1]]}
    ];
    ζ
  ], {Length@$M, $M}]

```

```

In[ ]:= {#, Z[#] // Simplify} & /@ AllKnots[{1, 7}] // MatrixForm

```

Get: ParentDirectory[File] in \$Path is not a string.

KnotTheory: Loading precomputed data in PD4Knots`.

Out[] // MatrixForm =

$$\begin{pmatrix} \text{Knot}[3, 1] & 1 \\ \text{Knot}[4, 1] & 1 \\ \text{Knot}[5, 1] & 1 \\ \text{Knot}[5, 2] & 1 \\ \text{Knot}[6, 1] & 1 \\ \text{Knot}[6, 2] & 1 \\ \text{Knot}[6, 3] & 1 \\ \text{Knot}[7, 1] & 1 \\ \text{Knot}[7, 2] & 1 \\ \text{Knot}[7, 3] & 1 \\ \text{Knot}[7, 4] & 1 \\ \text{Knot}[7, 5] & 1 \\ \text{Knot}[7, 6] & 1 \\ \text{Knot}[7, 7] & 1 \end{pmatrix}$$